

Chapter 5

Budget Formats and Performance

Government budgets are instruments for allocating scarce public resources among competing uses. The way a budget is organized affects expenditure control, managerial flexibility, the allocation of funds across programs, and the ability to evaluate whether public resources are being used efficiently. Budget formats therefore reflect different approaches to linking public expenditures with the services and outcomes produced by government.

This chapter examines the major approaches used to organize and evaluate government budgets. It begins with line-item budgeting, which emphasizes control over inputs, and then considers program and performance budgeting, which place greater emphasis on the allocation of resources across governmental activities and on the outputs and outcomes generated by those activities. The chapter also examines incremental and zero-based budgeting and compares the strengths and limitations of alternative budget formats for resource allocation and government performance. A brief overview of various budget types can be found on the webpage of the Georgia Municipal Association in a document titled [A Budget Guide for Georgia's Municipalities](#)).

5.1 Budget Philosophy

Budgets serve as a mechanism to protect, allocate, and control public resources. One fundamental requirement is efficiency, meaning that allocating \$1 to program A should provide the same level of benefit as allocating \$1 to program B. However, it can be difficult to value government goods and services due to several challenges. First, governments often have multiple objectives, some of which may conflict with each other. Additionally, there is no common mechanism to measure and compare the returns of different programs, making it difficult to assess which

investments are most beneficial. Another challenge is the variation in beneficiaries across programs, as well as differences between those who benefit from programs and those who pay for them. These factors complicate the process of evaluating budgetary allocations and ensuring that resources are distributed efficiently and fairly.

5.2 Budget Classification and Budget Types

Government budgets can be organized according to the stage of public production they emphasize. A useful framework distinguishes among inputs, outputs, and outcomes. These categories provide a basis for understanding the major budget formats used by governments. Inputs are the resources purchased and used by public organizations. They include labor, supplies, equipment, utilities, and capital assets. Inputs are generally the easiest components of government activity to measure because they are directly associated with expenditures. A budget organized primarily around inputs emphasizes financial control and accountability over how public funds are spent. Outputs are the goods and services produced by government agencies. Examples include the number of police patrols conducted, miles of roads maintained, library materials circulated, or applications processed. Outputs provide information about the quantity of services produced from available resources and therefore provide a closer connection between expenditures and government activity. And lastly, outcomes represent the broader results associated with government programs. Examples include lower crime rates, improved educational attainment, safer transportation systems, or improved public health. Outcomes are generally more difficult to measure than inputs or outputs because they are often influenced by factors outside the direct control of the government agency.

These distinctions are closely related to the major forms of budgeting. A line-item budget is primarily input-oriented and emphasizes control over expenditures such as salaries, supplies, and equipment. A program budget organizes expenditures according to programs or functions and therefore places greater emphasis on the purposes for which resources are used. A performance budget links expenditures to measures of activity, output, and, increasingly, outcomes in order to evaluate the efficiency and effectiveness of government programs. A zero-based budget differs somewhat from these classifications because its defining feature is the requirement that existing expenditures be periodically justified rather than automatically carried forward from a prior budget.

The distinctions among these budget formats are not mutually exclusive. Governments frequently combine elements of several approaches within a single budget system. Nevertheless, the progression from inputs to outputs and outcomes provides a useful framework for understanding how budget formats differ in the information they provide and in the decisions they are designed to support.

Operating Income	<i>Fiscal Year</i>		
	Last	Current	Next
Municipality	\$41,640	\$39,050	\$47,190
County	\$26,400	\$23,980	\$28,600
State	\$1,800	\$1,925	\$2,730
Donations	\$600	\$550	\$650
Fines and Other Income	\$1,560	\$2,310	\$2,730
Total	\$72,000	\$67,815	\$81,900

Table 5.1: Example budget of the operating income for a municipal library.

Operating Expenditures	<i>Fiscal Year</i>		
	Last	Current	Next
Salaries and Wages	\$28,980	\$27,170	\$33,345
Employee Benefits	\$20,100	\$18,700	\$21,255
Books and Periodicals	\$11,520	\$11,000	\$13,650
Staff Education	\$2,280	\$2,145	\$2,730
Telecommunications	\$1,710	\$1,650	\$2,048
Utilities	\$4,560	\$4,400	\$5,460
Equipment Repair	\$1,140	\$1,100	\$1,365
Supplies	\$1,710	\$1,650	\$2,048
Total	\$72,000	\$67,815	\$81,900

Table 5.2: Example budget of the operating income for a municipal library.

5.2.1 Line Item Budget

A traditional line-item budget focuses on listing expenditures by item, ensuring control and accountability over the purchase and use of inputs. It is input-oriented and classifies costs into general categories such as personal services, operating expenses, or capital outlays. The main focus is on resources purchased. Given how the line item budget is constructed, it does not allow for input substitution given, for examples, changes in input prices. We know from economics that when input prices change, there occurs a shift along the so-called isoquant. That is, producing the same output with a different combinations of inputs. We will see that the line item budget does not provide any information on activities or functions. Remember that (public) organizations exist to achieve goals. Consider the budget of a municipal library below. The budget listing the operating income and operating expenditures can be found in Tables 5.1 and 5.2.

The budget under consideration offers several advantages that facilitate financial management. It is easy to construct, making budgetary planning straightforward and manageable. Additionally, it is effective for maintaining cost control and

fiscal discipline, ensuring that expenditures stay within allocated limits. The transparency of this budgeting method also enhances accountability, making it easier to track and justify expenditures.

However, there are notable disadvantages. The budget primarily focuses on inputs rather than the outputs or outcomes, which does not provide a clear indication of the effectiveness or efficiency of spending. It also lacks insight into the broader purposes of expenditures, such as improvements in service programs, which can be crucial for strategic decision-making. For example, the line-item budget of the municipal library does not indicate an improvement of children's reading score in the community or how many books are checked out. Furthermore, its short-term, single-year focus does not support long-term planning and sustainability. Lastly, the rigidity of this budgeting method offers no flexibility for input substitution, restricting adaptability to changing needs or conditions.

5.2.2 Program Budget

Program budgets shift the focus of resource allocation from traditional line-items to major program areas with a goal-oriented framework, i.e., serving a common public purpose. Expenditures are listed by activities. This method prioritizes the expected results of services and activities, organizing budget allocations around a common purpose or function rather than rigid departmental structures. It typically adopts a multi-year perspective, offering greater flexibility in execution and necessitating a "crosswalk" to align program budgets with traditional line-item formats. Common program areas often include public safety, public works (e.g., infrastructure, roads), and human resources, which are aligned with the organization's overarching goals and often span multiple departments.

The features of program budgets include identifying products and services by their program structure, emphasizing outcomes over inputs. Programs with similar objectives are grouped together. This structure helps connect financial resources with program objectives through the use of crosswalks, enhancing the alignment between spending and strategic goals. Budgets are organized to group programs with similar objectives, such as fire and police departments under public safety, facilitating a unified approach to service provision. Moreover, by identifying cost and benefit data, program budgets enable the reallocation of resources to enhance program effectiveness and efficiency, supporting thorough reviews and adjustments of programs and subprograms.

The advantages of program budgets are manifold. They foster competition among alternative approaches to achieving public purposes, unbound by traditional organizational units. This budgeting method focuses on the government's provision of goods and services to society, rather than mere purchases, supporting a longer horizon for planning and decision-making.

However, program budgets are not without challenges. They require the use of crosswalks to bridge the gap with other budget classifications, often necessary

for accountability purposes. This approach can also disrupt legislative and administrative traditions, which typically focus on line-item appropriations, and the classification of expenditures into functions or programs can be complex, requiring significant effort to define clear goals and objectives. Additionally, the potential for overlapping structures among programs can complicate the budgeting process, presenting hurdles in clear accountability and execution. How to determine the programs (multiple competing objectives)

Below is an example of a possible program structure for a city. The example is taken from [Alan W. Steiss's lecture notes on program budgeting](#) at the University of Michigan:

1. **Public Safety:** (a) Law Enforcement, (b) Traffic Safety, (c) Fire Safety, (d) Maintenance of Public Order, (e) Prevention & Control of Other Hazards, and (f) Administration & Support
2. **Housing & Community Development:** (a) Housing Standards & Code Enforcement, (b) Community Improvement, and (c) Administration & Support
3. **Transportation:** (a) Traffic Control & Accessibility, (b) Street Development & Maintenance, (c) Mass Transportation, and (d) Administration & Support
4. **Environmental Enhancement & Protection:** (a) Environmental Health, (b) Water Services, (c) Sewer Services, (d) Sanitation Services, (e) Environmental Code Enforcement, and (f) Administration & Support
5. **Human Resources:** (a) Conservation of Health, (b) Personnel, Payroll & Benefits, (c) Vocational Rehabilitation, (d) Public Health Services, (e) Ambulance & Rescue Squad Services
6. **Education:** (a) Public Schools, (b) Adult & Vocations Education, (c) Community College Services, and (d) Higher Education Opportunities
7. **Recreation & Culture:** (a) Recreation & Parks, (b) Youth Opportunities, (c) Cultural Enrichment (inc. Libraries), and (d) Administration & Support
8. **Economic Development:** (a) Industrial Development & Promotion, (b) Job Opportunity Development, (c) Consumer Protection & Regulation, and (d) Administration & Support
9. **Finance & Revenue:** (a) Financial Operations (inc. Purchasing), (b) Assessment & Tax Collections, (c) Internal Audit & Records, (d) Recorder of Deeds, and (e) Administration & Support
10. **Executive Director & General Support:** (a) City Council, (b) Executive Management, (c) Budget and Control, (d) City Planning, (e) Human Relations & Affirmative Action, (f) Voter Registration & Elections, (g) Community Relations, (h) General Service Administration Support

5.2.3 Performance Budget

A performance budget links the allocation of public resources to the activities, outputs, and outcomes generated by government programs. Whereas a line-item budget emphasizes the purchase and control of inputs, a performance budget places emphasis on the relationship between expenditures and the services produced with those resources. The objective is to provide information that can be used to assess the efficiency and effectiveness of government programs and to improve managerial accountability. Performance budgeting therefore requires the development of measures that describe both the volume of government activity and the results of that activity. These measures generally fall into three categories:

- **Workload measures** describe the quantity of activities undertaken or services provided. Examples include the number of customers served, miles of streets maintained, inspections completed, or emergency calls answered.
- **Efficiency measures** relate outputs to the resources used to produce them. A common efficiency measure is cost per unit of output, such as the cost per inspection or cost per emergency response.
- **Effectiveness measures** assess the extent to which a program achieves its objectives. Examples include response times, customer satisfaction, the percentage of targets achieved, or changes in conditions that the program is intended to influence.

The distinction between efficiency and effectiveness is particularly important. A program may produce a given service at relatively low cost and therefore be technically efficient, while still having little effect on the outcome it is intended to improve. Conversely, a program may be effective in achieving its objectives but require substantial resources. Performance budgeting attempts to provide information on both dimensions rather than evaluating a program solely on the amount it spends.

Consider the public safety programs of the [City of Sunnyvale, California](#). The city's budget includes workload measures such as the number of traffic enforcement stops, parking and traffic citations issued, traffic complaints addressed, and police responses to emergency events. It also reports performance measures such as average police response times to emergency and urgent events. These indicators provide information beyond the amount spent on police services by linking expenditures to the level and quality of services provided.

For performance information to be useful in budgeting, the measures themselves must be carefully designed. Measures should be clearly related to program objectives, sufficiently specific to permit meaningful evaluation, observable or measurable, and economical to collect. Measures selected merely because data are readily available may provide little useful information about program performance. The number of performance measures should also remain manageable, because a large set of indicators can make it difficult to identify which results are most important.

Performance budgeting can improve internal management by providing information about service levels, unit costs, productivity, and program results. It can also strengthen accountability by making the relationship between public expenditures and government performance more transparent. Managers may be given greater flexibility in the use of inputs when they are held accountable for achieving specified performance objectives.

Performance budgeting also has important limitations. Developing and maintaining performance measures can be costly, and many government outcomes are difficult to measure. Outcomes may also depend on economic, demographic, or social conditions that are outside the control of the agency. Comparisons across programs can be difficult because different programs produce fundamentally different services and outcomes. In addition, the selection of performance measures can influence managerial behavior, creating incentives to concentrate on measured activities while giving less attention to objectives that are more difficult to quantify.

Performance budgets and program budgets are closely related but emphasize different aspects of resource allocation. Program budgeting organizes expenditures around common governmental purposes or programs, while performance budgeting provides information about the efficiency and effectiveness with which those programs operate. In practice, the two approaches are frequently used together, with program structures defining the purposes of government spending and performance measures providing information about the results generated by that spending.

5.2.4 Good Performance Measures

The usefulness of a performance budget depends heavily on the quality of the measures used to evaluate government activities. A performance measure should provide information that is relevant to the objectives of the program and useful for evaluating how resources are converted into public services and outcomes. Measures should not be selected simply because the underlying data are readily available.

Good performance measures generally have several characteristics. They should be relevant, meaning that they are closely connected to the objectives of the program; clearly defined, so that changes in the measure have an unambiguous interpretation; and measurable, so that performance can be evaluated consistently over time. Measures should also be economical, since the cost of collecting and maintaining performance information should not exceed its value for budgeting and management.

Performance measures should also reflect outcomes that are reasonably related to the activities of the agency. This is particularly important for effectiveness measures. Many government outcomes are influenced by economic, demographic, and social factors outside the control of a particular program. A decline in crime, for example, may be an important public safety outcome, but changes in crime

cannot necessarily be attributed entirely to police expenditures. For this reason, performance systems often combine measures of outputs, service quality, and outcomes rather than relying on a single indicator.

Finally, a useful set of performance measures should be limited and balanced. Too many measures can obscure rather than clarify program performance, while reliance on a single measure may provide an incomplete picture and create incentives for managers to improve the measured indicator at the expense of other objectives. A small set of complementary measures can provide a more informative assessment of efficiency and effectiveness.

5.2.5 The Relationship Between Program and Performance Budgeting

Program and performance budgeting are closely related because both move beyond the input-oriented focus of the traditional line-item budget. However, they emphasize different aspects of the allocation and use of public resources. A program budget organizes expenditures around programs or functions that serve a common public purpose. Its primary contribution is to show how resources are allocated among competing governmental objectives rather than among organizational units or categories of inputs.

A performance budget builds on this program structure by linking the resources devoted to a program with measures of the activities, outputs, and outcomes generated by that program. Performance information therefore provides an additional basis for evaluating how effectively and efficiently resources assigned to programs are being used. In economic terms, program budgeting is primarily concerned with the allocation of resources across purposes, while performance budgeting provides information about the productivity and effectiveness of resources within those purposes.

The two approaches are therefore complementary rather than competing. A government may first organize expenditures into programs and then attach workload, efficiency, and effectiveness measures to each program. For example, a public safety program may combine police, fire, and emergency services within a common program structure. The program budget identifies the resources devoted to public safety, while the performance component reports measures such as emergency responses, response times, costs per response, and other indicators of service performance.

The distinction is important because information about performance does not by itself determine how resources should be allocated across programs. A highly efficient program may still receive fewer resources if policymakers place greater value on the objectives of another program. Performance measures inform budget decisions, but the allocation of resources among competing public purposes ultimately also depends on the relative social benefits and priorities attached to those purposes.

5.2.6 Zero-Based Budgeting

Most government budgets are prepared incrementally. Under incremental budgeting, the current or previous year's budget serves as the starting point for the next budget. Attention is therefore concentrated primarily on proposed increases or reductions rather than on reconsidering the entire level of spending. This approach reflects the fact that many government programs, staffing levels, and service commitments continue from year to year.

Incremental budgeting has several practical advantages. It reduces the administrative burden of preparing the budget, provides continuity in government operations, and allows policymakers to concentrate on changes at the margin. At the same time, it can create path dependence because existing expenditures become embedded in the budget base. Programs may continue to receive funding largely because they were funded previously, even when priorities or economic conditions have changed.

Zero-based budgeting was developed as an alternative to this incremental approach. Under zero-based budgeting, existing expenditures do not automatically receive priority simply because they are part of the current budget. Instead, agencies are expected to periodically justify their activities and expenditures and to compare them with alternative uses of public resources. The analytical focus therefore shifts from changes in the budget to the justification of the budget itself.

In principle, zero-based budgeting requires agencies to identify activities, estimate the resources required to provide them, and evaluate alternative levels or methods of service delivery. Activities can then be ranked according to their relative importance or contribution to organizational objectives. This process is intended to make existing spending compete more directly with proposed new spending and to encourage a more systematic consideration of priorities.

The principal advantage of zero-based budgeting is that it subjects existing programs to greater scrutiny. It can reveal activities that are no longer priorities, encourage consideration of alternative methods of service delivery, and generate detailed information on program costs and operations. It may therefore improve the allocation of resources when historical spending patterns no longer correspond closely to current priorities.

The major limitation is the substantial amount of information and administrative effort required. Government agencies often operate programs that cannot realistically be eliminated or reduced to zero, and evaluating every activity each budget cycle can impose large costs. In practice, governments therefore rarely implement a completely zero-based process. Elements of zero-based budgeting are more commonly used selectively, particularly when programs are being reviewed or when fiscal pressures require significant changes in spending priorities.

Full ZBB is uncommon mainly because the information and decision costs are extremely high. A genuine zero-based process would require agencies to

repeatedly justify virtually every activity, develop alternative service levels, estimate the costs and consequences of those alternatives, and rank competing activities across the organization. For a large government, the number of decisions quickly becomes enormous, while policymakers have limited time and information to evaluate them.

There are also structural reasons. Much government spending is constrained by law, contracts, debt obligations, personnel commitments, or continuing public-service responsibilities, so a true “zero” starting point is often unrealistic. In addition, existing programs embody accumulated investments, organizational capacity, and political commitments. Reconsidering all of them from scratch each year would sacrifice much of the continuity that makes government operations possible. As a result, governments more often use selective or periodic zero-based reviews within an otherwise incremental budget process, concentrating intensive analysis on particular programs, agencies, or areas of spending rather than rebuilding the entire budget from zero each year.

5.2.7 Comparing and Choosing Budget Formats

Each budget format emphasizes a different aspect of public resource allocation. A line-item budget provides strong control over inputs and makes it relatively easy to determine how public funds are spent. A program budget shifts attention toward the purposes for which resources are allocated, while a performance budget adds information on the efficiency and effectiveness with which programs operate. Zero-based budgeting focuses more directly on whether existing expenditures continue to be justified relative to alternative uses of public resources.

No single format is superior in all circumstances. The appropriate approach depends on the information needs of policymakers and managers. Strong expenditure control may favor line-item budgeting, while decisions about the allocation of resources across policy objectives benefit from a program structure. Performance information is most useful when managers and policymakers need to evaluate service levels, productivity, or program results. Zero-based techniques are particularly useful when existing spending patterns need to be reconsidered or when fiscal constraints require a reassessment of priorities.

In practice, governments often combine elements of several approaches. A budget may retain line-item classifications for legal control and accounting purposes, organize expenditures by program for planning and allocation, and include performance measures to evaluate results. Periodic zero-based reviews may then be used to examine selected programs or areas of spending. The choice of budget format is therefore less a matter of selecting one exclusive system than of combining budgeting tools in a way that supports fiscal control, efficient resource allocation, and effective government performance.

Program	Current Budget (\$ million)	a_i	α_i	Policy Weight
Police	5.0	10,000	0.70	0.45
Fire	3.0	5,000	0.60	0.35
Library	2.0	80,000	0.50	0.20

Table 5.3: Parameters for the exercise *Performance-Based Budget Allocation*

Program	Expenditure		Output		Outcome	
	Year 1	Year 2	Year 1	Year 2	Year 1	Year 2
Police	20.0	22.0	18,000	19,500	42	39
Fire	12.0	13.0	5,000	5,150	7.2	6.8
Library	6.0	6.5	180,000	205,000	68	74

Table 5.4: Expenditures are reported in millions of dollars. Police and fire outputs are measured as the number of responses, while library output is measured as the number of visits. Police outcomes are measured by the crime rate, fire outcomes by average response time in minutes, and library outcomes by the percentage completing the literacy program.

5.3 Exercises

1. **Performance-Based Budget Allocation (***)**: A city currently allocates \$10 million among three programs: Police, Fire, and Libraries. The city has received an additional \$1 million from the state and must decide how to allocate it. The objective is to use performance information to allocate the additional funds while recognizing diminishing returns to spending. Consider the information in Table 5.3. For each program, output is determined by the following equation

$$Q_i = a_i \cdot B_i^{\alpha_i}$$

where Q_i is program output and B_i is the program budget in millions of dollars. Because $0 < \alpha_i < 1$, each program exhibits diminishing marginal returns to additional spending.

- a. Calculate the baseline output for each program using its current budget.
- b. Calculate the additional output generated by increasing each program's budget by \$0.1 million.
- c. The city must allocate the additional \$1 million among the three programs in increments of \$0.1 million. The allocations must satisfy $\Delta B_P + \Delta B_F + \Delta B_L = 1$.
- d. For each proposed allocation, calculate the new output of each program.

- e. Calculate the following weighted performance index for which a value of 100 represents the performance associated with the original budget.

$$PI = 100 \cdot \sum_i w_i \left(\frac{Q_{i,\text{new}}}{Q_{i,\text{baseline}}} \right)$$

- f. Determine the allocation of the additional \$1 million that maximizes the performance index.
- g. Explain why the allocation that maximizes the performance index does not necessarily assign the additional funds to the program with the largest total output.
2. **Measurement Classification** (**): Choose a public or campus service specific to Indianapolis that you personally rely on regularly, such as a bus route, a library, a dining hall, or a recycling program. Design one workload measure, one efficiency measure, and one effectiveness measure for that service. Define each precisely enough that two different people applying your definition would arrive at the same number. Explain, in your own words, why your workload measure by itself does not tell you whether the service is effective. Then describe one specific action that a manager of the service could take to improve your efficiency measure without making the service any better, and use that example to explain why the choice of performance measures can create incentives that work against a program's underlying objectives.
3. **Evaluating Government Performance** (**): A city operates three public programs. Table 5.4 reports each program's expenditures and selected performance measures for two consecutive years.
- a. Calculate the percentage change in expenditures and outputs for each program.
- b. Calculate an average cost per unit of output for each program in both years. For example:

$$\text{Cost per unit of output} = \frac{\text{Program Expenditure}}{\text{Program Output}}$$

- Calculate the percentage change in unit cost between the two years.
- c. Based only on the expenditure and output data, identify which program experienced the largest improvement in **technical efficiency**. Explain the basis for your conclusion.
- d. Evaluate the outcome measures reported for each program. Classify each measure as relatively more or less under the control of the agency and explain your classification.
- e. Suppose the following additional information becomes available. First, the unemployment in the city declined substantially between Year 1 and Year 2. Second, the fire department replaced several older vehicles during Year 2. And lastly, the library introduced an online

- reservation system and extended weekend hours. Reconsider your interpretation of the observed performance changes. Identify which improvements can reasonably be attributed to agency performance and which may partly reflect external conditions or changes in inputs.
- f. The city manager proposes allocating next year's budget increases entirely according to the percentage improvement in each program's reported outcome measure. Evaluate this proposal. Discuss at least two problems with using the outcome measures in this way.
 - g. Recommend one workload measure, one efficiency measure, and one effectiveness measure for each program. Briefly justify each measure.
 - h. Prepare a short recommendation to the city council identifying which program appears to have improved its performance the most. Your recommendation must distinguish explicitly among changes in inputs, outputs, efficiency, and outcomes, and it must identify any conclusions that cannot be supported by the available data.